

DL Routing Performance in DECT-2020 NR Massive Mesh Networks

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Abstract—DECT-2020 NR, a global radio communication standard developed by ETSI, is designed to support industrial Internet of Things (IoT) applications. It complies with the requirements set by the ITU IMT-2020 framework. This study presents comprehensive simulations to evaluate downlink (DL) packet-forwarding performance in dense mesh networks using the DECT-2020 NR standard. Several studies address uplink performance; consequently, this specific downlink-focused research question has received limited attention in prior academic work. The study focuses on two routing mechanisms specified in the standard: Selective Flooding and Selective Source Routing, evaluated in a 1,200-node, 10-hop, large-scale deep mesh network. The simulation results indicate that the downlink (DL) capacity for Selective Flooding—where no routing information is required—is approximately 6 packets per second. For Selective Source Routing, the downlink (DL) throughput increases with the number of deterministic routing hops, reaching close to 145 packets per second when the system uses only a single 1.728 MHz channel. Both capacities are achieved while keeping the packet outage rate below 1% to comply with the IMT-2020 requirements.

Index Terms—DECT-2020 NR, selective flooding, selective source routing, IoT, massive mesh networks

I. INTRODUCTION AND MOTIVATION

The design objective of DECT-2020 NR was to create a wireless communication technology that can be deployed and operated by individuals or companies anywhere and combine mesh communication support with modern OFDM-based radio access [1]. The main target applications during the design phase were massive machine-type communication (mMTC) and ultra-reliable low-latency communication (URLLC), as defined by ITU-R for IMT-2020 (i.e., 5G) [2]. IMT-2020 technologies supporting the mMTC service type are required to support high IoT device density (1 million devices per km²), where each device generates a 32-byte data packet periodically every 2 hours [3], [4].

After significant evaluation efforts (e.g., [5]), DECT-2020 NR was accepted as part of the ITU-R recommendation for radio interfaces of International Mobile Telecommunications-2020 (IMT-2020), i.e., 5G [6].

Several publications, such as [7], [8], provide extensive analyses of uplink performance for mMTC services. In [9], the authors analyse and present a large number of potential future use cases that would benefit from DECT-2020 NR wireless communication. When considering these use cases, it is apparent that, in many modern Internet of Things (IoT) applications, the data traffic is not only uplink-oriented, but

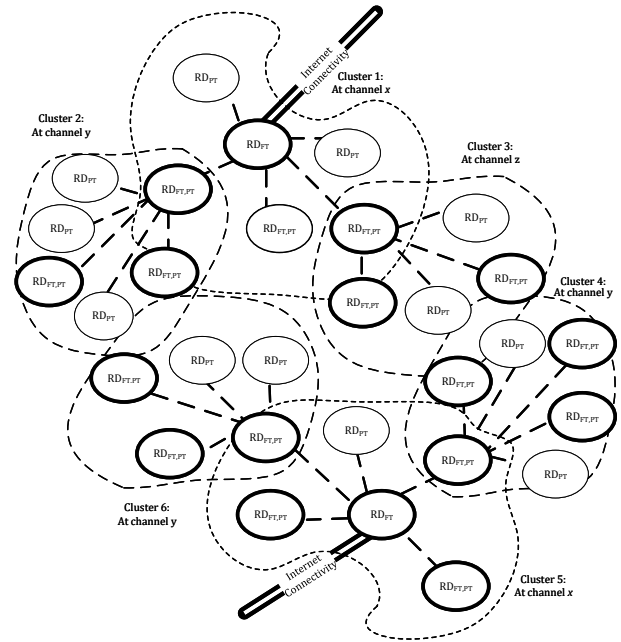


Fig. 1. Illustration of DECT-2020 NR Mesh network structure [10]

downlink traffic (i.e., from cloud to device) also plays an important role.

Therefore, we focus on how DECT-2020 NR supports downlink traffic in massive IoT deployments, using technologies available in DECT-2020 Release 1 and Release 2.

The rest of this paper is structured as follows. Section II reviews related work and discusses challenges in downlink routing. Section III describes the physical and MAC layers of DECT-2020 NR and explains how routing is designed in DECT-2020 NR. Section IV presents the simulation scenario and the strategy used to assess the downlink performance of DECT-2020 NR. Section V discusses the simulation results and highlights key findings. Finally, Section VI concludes the paper by summarizing the findings.

II. STATE OF THE ART AND CHALLENGES

Several IoT technologies operating in the unlicensed spectrum—such as BLE [11], LoRa [12], ZigBee [13], and Wi-SUN [14]—are commonly available in the market. Among these technologies, BLE mesh communication uses a managed-flooding approach, where a time-to-live (TTL) field limits the

number of hops a packet can traverse. Although managed flooding can be inefficient in many scenarios, it remains a practical choice for small networks with low throughput requirements. LoRa employs a proprietary physical layer, but no standardized multi-hop mesh routing protocol has been defined for it so far. Wi-SUN primarily adopts source routing [15], [16] for downlink traffic and relies on RPL (Routing Protocol for Low-Power and Lossy Networks) as its IPv6 packet-forwarding mechanism. The ZigBee Alliance [13] defines standards for both the network and application layers, and ZigBee uses a hybrid approach for downlink routing, combining many-to-one routing with source routing.

The research community has examined a wide range of IoT technologies—including BLE [17], [18], LoRa [19]–[21], Wi-SUN [22]–[24], DECT-2020 NR mesh [25], ZigBee [26], [27], and IEEE 802.11s [28], [29]—covering various aspects of system performance. In this context, downlink routing in multi-hop IoT mesh networks involves numerous challenges, including route discovery [30], [31], reliability [24], [32], energy efficiency [28], [33], and—critically—network capacity and scalability [32]. This is further intensified by the fact that, unlike traditional computer networks, IoT nodes operate with limited energy, memory, and computational power.

These challenges also grow significantly with increasing network size, higher node densities, and deeper hop counts. The research on existing technologies has largely been limited to small-scale networks, typically around a hundred nodes or fewer. However, the need for IoT deployments to operate at a much larger scale is becoming unavoidable. Even today's systems are expected to support massive node densities, deeper multi-hop paths, and larger coverage areas than existing technologies have demonstrated. The DECT-2020 NR standard [10] has already committed to addressing these large-scale requirements, as mentioned in Section I. In this context, this paper investigates the downlink capacity of a DECT-2020 NR mesh network in terms of delay and throughput, evaluating two routing strategies—Selective Flooding and Selective Source Routing—across a large-scale mesh comprising 1,200 nodes and more than 10 hops.

III. DECT-2020 NR AND DL ROUTING

A. PHY and MAC

The DECT core band of interest in this study in Europe ranges from 1880 MHz to 1900 MHz and supports 11 channels, each with a bandwidth of 1.728 MHz, for DECT-2020 NR. The DECT-2020 NR physical layer (PHY) uses a 10ms frame structure that is divided into 48 subslots when operating with a 1.728 MHz channel bandwidth [34], [35]. OFDM was selected for DECT-2020 NR because it provides robust state-of-the-art modulation and can be combined with modern channel coding, Turbo coding, HARQ, and MIMO technologies. Furthermore, standard CP-OFDM provides efficient channel estimation and equalization at the receiver. The physical-layer packet structure is identical for all transmissions, enabling a single PHY processing chain at the transmitter and receiver [35].

On top of the physical layer is the Medium Access Control (MAC) protocol, which forms cluster-tree topologies (see Fig. 1). This is possible because radio devices (RDs) can operate in both Fixed Termination (FT) and Portable Termination (PT) modes simultaneously (i.e., as $RD_{FT,PT}$). An RD that operates only in FT mode (RD_{FT}) provides Internet access to the DECT-2020 NR network, whereas an $RD_{FT,PT}$ operates as a router and an RD_{PT} operates as a simple leaf device. The main FT functionality of both RD_{FT} and $RD_{FT,PT}$ is to control local radio resources and provide connectivity for their associated RDs, whereas the main PT functionality of both $RD_{FT,PT}$ and RD_{PT} is to select an FT for association and operate based on instructions received from the FT device.

DECT-2020 NR MAC supports two types of access methods: Scheduled Access and Random Access, the are under control of FT functionality RD_{FT} and $RD_{FT,PT}$. The latter access method used in this study. Before transmitting on the Random Access Channel (RACH), an RD in mode (RD_{PT}) must obtain RACH resources from an FT functionality of (RD_{FT}) or ($RD_{FT,PT}$). The RD PT mode then performs a channel assessment to determine whether the channel is busy. If the channel is not busy, the RD initiates transmission. Conversely, if the channel is busy—as indicated by the detection of energy from other transmissions—the RD waits for a random backoff time before re-assessing the channel to attempt transmission; that is, the RD uses the Listen Before Talk (LBT) protocol.

In DECT-2020 NR, the same principles can also be used for downlink transmission, i.e., from RD_{FT} towards $RD_{FT,PT}$ and RD_{PT} . In this case, the $RD_{FT,PT}$, or RD_{PT} communicates its random-access resources (operating frequency, time, and other RACH parameters) to its parent RD_{FT} or $RD_{FT,PT}$. Then, an RD_{FT} or $RD_{FT,PT}$ performs random-access transmission towards the RD to enable downlink transmission. This configuration is the main method used in the ETSI DECT-2020 NR Smart Metering, City and Buildings Access Profile [36] for downlink transmissions.

B. DLC and Routing

The DLC (Data Link Control) protocol layer provides several operating modes: transparent mode, segmentation mode, ARQ (Automatic Repeat Request) mode, and a combined mode that incorporates both segmentation and ARQ functionalities [37]. Above these, the routing service (see Fig. 2) supports packet routing for uplink, downlink, and device-to-device communications. Routing uses long RD addresses as the source and destination identifiers, providing unique addressing within a single network. Uplink routing is based on a clustered-tree topology formed during the association process. Each PT functionality of the RD periodically searches for and assesses neighboring FT devices, i.e., RD_{FT} and $RD_{FT,PT}$, and their route-cost values towards the backend to select an optimal parent. The route cost is obtained from the Route Info IE, which all RD_{FT} and $RD_{FT,PT}$ nodes include in Network and Cluster Beacon messages.

In Release 1, downlink routing is based on Selective Flooding within the clustered-tree topology. The RD_{FT} forwards

a routing data PDU to all associated $RD_{FT,PT}$ and the distribution tree is pruned whenever possible until the final destination is reached. Because routing decisions do not rely on routing tables, there is no routing-maintenance signaling and no memory requirement for storing routes; consequently, the network size is not restricted by the number of devices under a single RD_{FT} [37]. Furthermore, the solution is robust: since packets are flooded, topology changes (e.g., re-association to another RD_{FT}) do not necessarily cause downlink packet loss while routes converge. However, selective flooding introduces unnecessary transmissions, which limits the downlink capacity.

Separating routing methods for each traffic direction ensures that each direction can be independently developed and optimized. This was realized in DECT-2020 NR Release 2, where downlink routing was enhanced with Selective Source Routing and Cached Routing to improve downlink capacity. [37]

In generic source routing, each device sends a registration message (including the device's parent-router address) to the RD_{FT} with Internet access. The RD_{FT} then constructs downlink paths for all devices using the received registration messages. When a downlink data packet is transmitted, the RD_{FT} includes the sequence of addresses along the path to the destination device. As a result, association changes must be reported to the RD_{FT} , and the downlink data PDU overhead increases due to the embedded path addresses. Because these factors limit scalability in large-scale deployments, DECT-2020 NR introduces a mechanism that allows the system to control how many hops from the RD_{FT} use source routing; if re-associations do not affect the portion of the path where source routing is used, no re-registration is needed. For example, if source routing is used only for the first hop (from the RD_{FT} to its member $RD_{FT,PT}$), route changes deeper in the tree do not require re-registration, significantly reducing registration signaling.

From the above discussion, the basic difference between these two routing strategies can be explained with reference to Fig. 1. Let us assume that some RD_{PT} of cluster2 is the ultimate destination of a DL packet. In source routing, as the next routing hop is embedded in the DL packet, the RD_{FT} of cluster 1 will forward the packet directly to the router $RD_{FT,PT}$ of cluster 2. However, in selective flooding, no next hop information is known beforehand for a router; hence, if none of its cluster members is the ultimate destination, the router unicasts the packet to all routing members within the cluster for further route exploration. Under this strategy, the RD_{FT} of cluster 1 will unicast the packet to the $RD_{FT,PT}$ of cluster 2 and cluster 3, even though cluster 3 is not the correct direction to explore. As a result, if the depth of the network tree is large, such flooding will generate a high volume of redundant packets.

With Cached Routing (also introduced in Release 2), any routing device ($RD_{FT,PT}$) may learn and store the correct downlink path from observed uplink packets. This requires each $RD_{FT,PT}$ to store learned RD addresses for its member RD_{PT} , $RD_{FT,PT}$ nodes, but the storage requirement

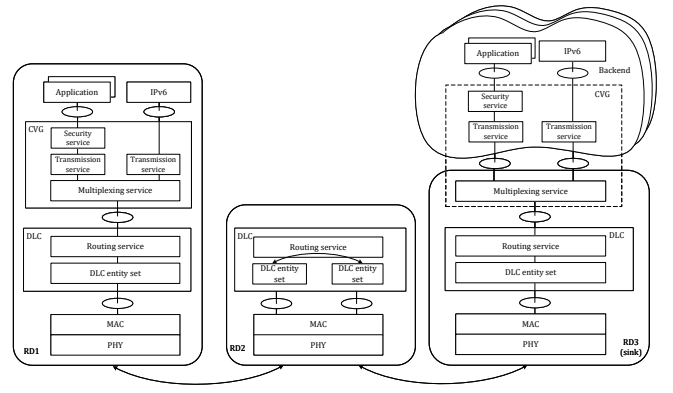


Fig. 2. DECT-2020 NR Protocol Architecture [37]

decreases toward the leaves of the tree. If source-routing information is missing from a received PDU, or if an $RD_{FT,PT}$ does not have valid cached-routing information, it falls back to the Release 1 Selective-Flooding procedure.

IV. PERFORMANCE ANALYSIS

In this study, two downlink routing approaches are explored for dense multi-hop DECT-2020 NR mesh networks: Selective Flooding and Selective Source Routing. To assess overall system performance, DECT-2020 NR protocol features are incorporated into the ns-3 simulator, an open-source, high-speed, event-driven framework written in C++ and widely adopted in network research.

A. Experimental setup

Deployment scenario. The RDs are deployed randomly in a circular region with a radius of 700 m, representing a mixed indoor–outdoor environment of roughly 1.5 km². A single sink node (i.e., RD_{FT}) is placed at the centre. The network includes 1,200 RDs, with 80% located indoors—20% in high-loss buildings and 60% in low-loss buildings—while the remaining 20% are deployed outdoors. The remaining simulation parameters are presented in Table I. The simulation experiments employ the NIST Device-to-Device (D2D) channel model [38].

The simulation outcomes are based on 50 independent random deployments. Of the 1,200 RDs, roughly 360 nodes (on average) act as routers, i.e., $RD_{FT,PT}$, and are responsible for forwarding tasks. The highest hop count is, on average, 10 hops. All communication uses a single channel with a transmit power of 19 dBm.

Network Formation. The deployed RDs autonomously form a tree-based topology according to the association rules. To join a router, each RD evaluates a cost function that incorporates factors such as hop distance to the sink, minimum received signal quality from the router, and other relevant metrics. Sink and every router periodically broadcasts a beacon that includes association-related parameters. In this simulation setup, each cluster can include up to 14 member RDs.

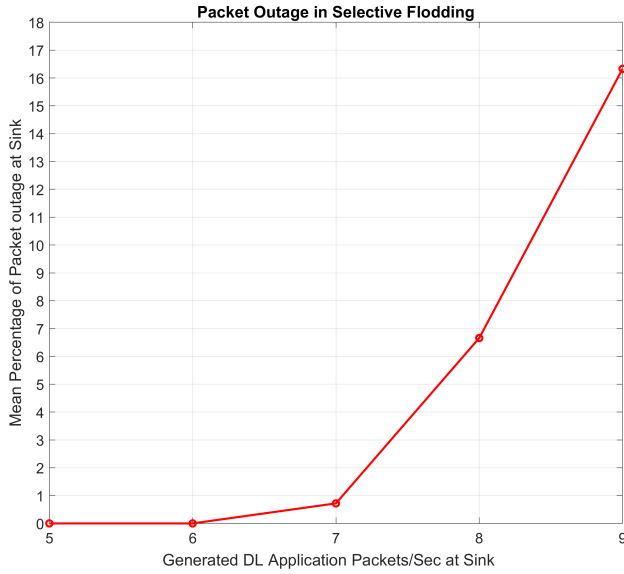


Fig. 3. Mean DL Packet outage percentage (Selective Flooding)

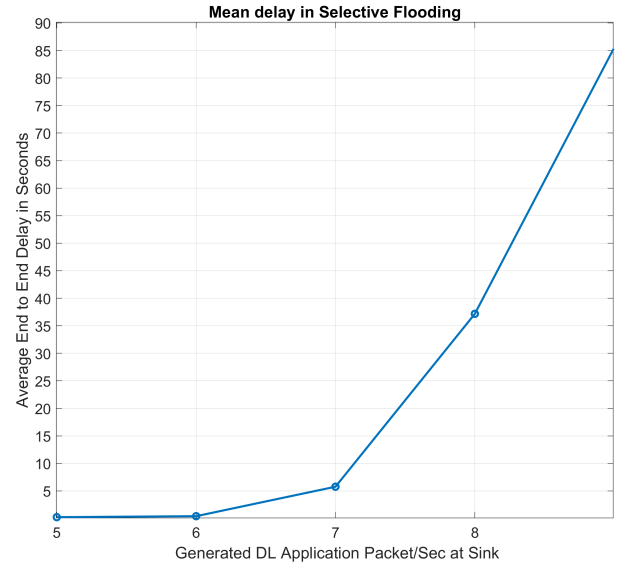


Fig. 4. Mean DL Packet delay in seconds (Selective flooding)

In a single random realization, the resulting tree structure after the association remains the same for the entire simulation because no re-association is assumed once all RD have associated with the parent RD, RD_{FT} or $RD_{FT,PT}$. However, for the set of 50 random scenarios, the network topology is generated independently for each run, producing a random topology every time.

Traffic Model. The sink, RD_{FT} , creates DL packets for each node at an equal rate, following a Poisson arrival pattern. The transmission of Each MAC PDU (Protocol Data Unit) occupies 3 slots, corresponding to 6 sub-slots, and carries a payload of roughly 1,352 bits (approximately 169 bytes). QPSK modulation with a 1/2 coding rate is used for transmission, matching Modulation and Coding Scheme 1 (MCS 1).

PHY and MAC Parameters. The PHY and MAC layer parameters used in this experiments are summarized in Table I. The MAC utilized the Random Access transmission method as discussed in Section III-A.

B. Numerical results

1) Selective Flooding: In this experiments, when a router (RD_{FT} , or $RD_{FT,PT}$) receives a packet, it first checks whether the destination matches any of its associated nodes. If it does, the packet is delivered solely to that specific node. If not, the router sends the packet individually to each of its routing member RD $RD_{FT,PT}$ using separate unicast transmissions. For instance, in this latter situation, if the router has 8 routing members, it will send the same packet 8 times—once to each member router.

To evaluate performance, the downlink (DL) application packet rate at the sink RD_{FT} is gradually increased, and the resulting packet outage and delay are measured. The mean packet outage and mean delay are shown in Fig. 3 and Fig. 4,

respectively. In Fig. 3, the mean packet outage stays at 0% until the application packet rate reaches 6 packets/s (pps). After that, it increases quickly: at 7 packets/s it is 0.72%, and at 8 packets/s it rises to 6.66%.

Similarly, Fig. 4 shows that the mean delay starts increasing once the application packet rate goes above 6 packets/s. The mean delay become 0.42 s, 4.77 s, and 37.17 s for 6, 7, and 8 packets/s, respectively. From Fig. 3 and Fig. 4, it is clear that for this DL flooding scheme, the system remains stable until the application packet rate reaches around 6 packets/s. Beyond this rate, both the mean packet outage and the end-to-end delay grow rapidly.

2) Selective Source Routing: In this experiment, each uplink (UL) data packet – or a small dedicated route-information packet – first carries the RD's registration information, as discussed in section III-B to the RD_{FT} . The RD_{FT} constructs the routing information for downlink path. During downlink transmission, the stored route of the destination node is inserted into the corresponding DL packet. This embedded information allows intermediate routers to forward the DL packet correctly along the intended path to its final destination. [37]

The mean delay and packet-outage percentage are evaluated for this scheme. Unlike selective flooding, source routing paths are predetermined; the route of each downlink packet is fully defined in advance, eliminating any uncertainty in its delivery path. The number of hops that forward a packet deterministically—without flooding—is determined by the information element (IE) included in the packet header. The delay-capacity performance for 1-hop source routing and end-to-end source routing is illustrated in subsections (a) and (b), respectively.

a) Only First-hop Selective Source Routing: In this method, selective source routing is applied only at the first

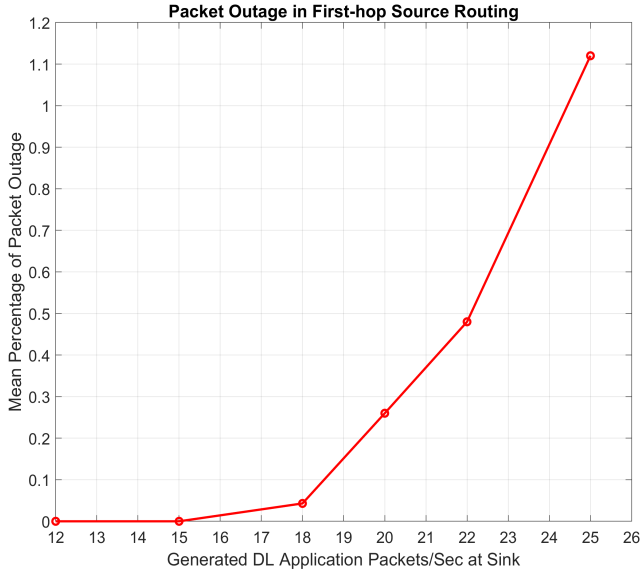


Fig. 5. Mean DL packet outage percentage (first-hop Selective Source Routing)

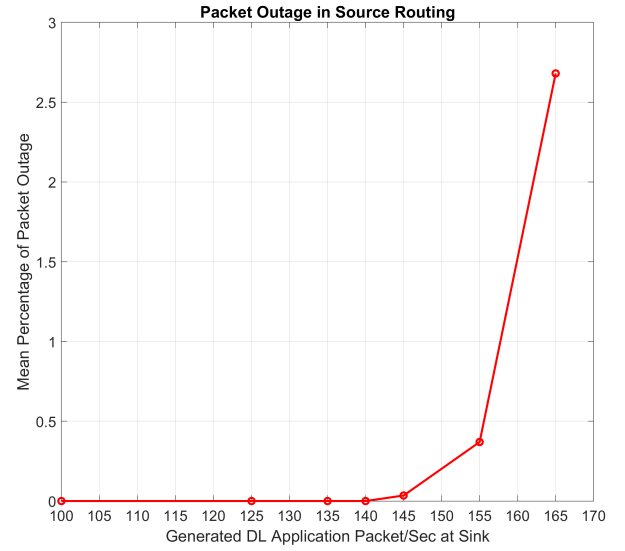


Fig. 7. Mean DL Packet outage percentage (end-to-end Source Routing)

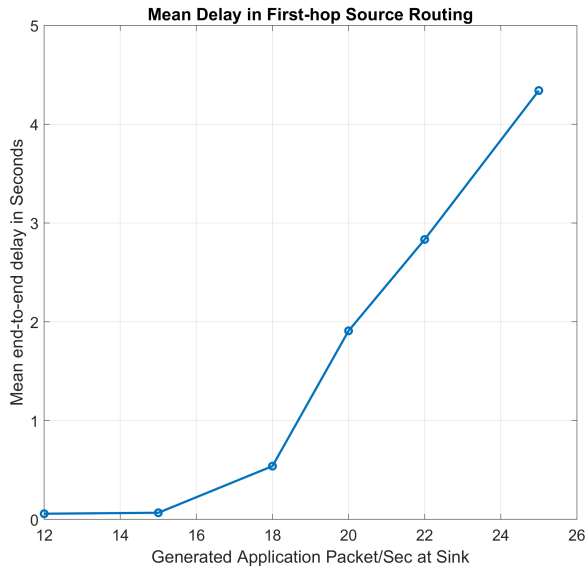


Fig. 6. Mean DL packet delay in seconds (first-hop Selective Source Routing)

hop after the sink RD_{FT} . When the sink sends a downlink packet, it forwards the packet along the correct branch, as described earlier. After that first hop, the intermediate routers use Selective Flooding until the packet reaches its destination. The benefit of this method is that actual DL PDU does not need to include any additional address information and thus protocol overhead is not affected.

In Fig. 5, the outage stays at zero up to 15 packets/s. It then increases to 0.043 at 18 packets/s and rises to 0.26 and 0.48 at 20 and 22 packets/s. In Fig. 6, the mean delay is 0.067 s, 0.54 s, 1.9 s, and 2.83 s at 15, 18, 20, and 22 packets/s respectively.

Both figures show that outage and delay grow quickly after 18 packets/s indicating that the system remains stable up to 18 packets/s.

b) End-to-End Source Routing: In contrast to first-hop deterministic routing, end-to-end source routing forwards the packet deterministically all the way to the final hop of the network, with no flooding occurring anywhere in the network. In Fig. 7 the packet outage percentage begins to increase rapidly at around 145 packets/s. The mean packet outage percentages are 4.77×10^{-5} , 3.4×10^{-2} , 3.72×10^{-1} , and 2.68, corresponding to application packet rates of 140, 145, 155, and 165 packets/s, respectively.

The same tendency is observed in Fig. 8. The mean delays at 140, 145, 155, and 165 packets/s are 0.04 s, 0.42 s, 3.98 s, and 22.15 s, respectively. As the application-packet rate approaches 145, the mean delay begins to grow without bound. Figures 7 and 8 clearly indicate that the network DL capacity reaches close to 145 packets/s at steady state. It is worth noting that in our experiments, a dense multihop network of similar size reached an uplink (UL) capacity of around 145 packets per second also in a single channel.

In addition to DL first-hop source and end-to-end source routing, extensive simulations were carried out to examine how performance changed as the deterministic hop count from sink was gradually increased. The summary in Table II shows how the network's overall downlink (DL) capacity changes as it moves from selective flooding to source routing for a growing number of selective source routing hops.

Figures 3–8 and Table II show that increasing the selective source routing hops improves both capacity and delay performance. This happens because predefined routes reduce redundant flooding packets in the network. These excess flooding packets would otherwise create heavy queueing delay

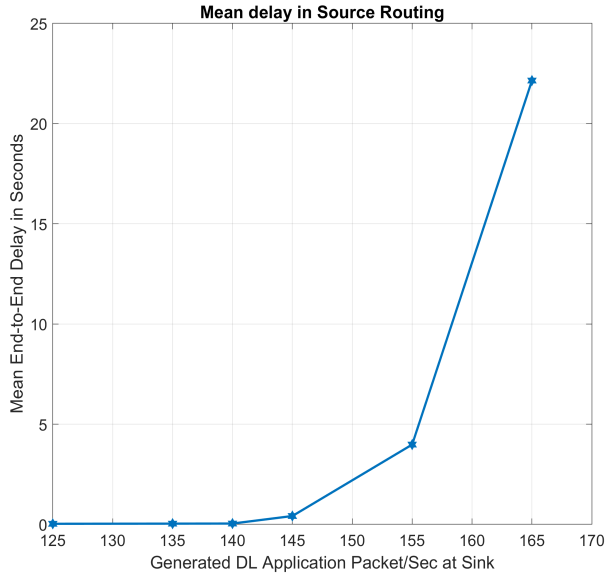


Fig. 8. Mean DL Packet delay in seconds (end-to-end Source Routing)

TABLE I
CONSIDERED SYSTEM SIMULATION PARAMETERS

Parameter	Value
DECT-2020 NR FT	1, height 5 m, outdoors
Carrier frequency	1900 MHz
Number of channels	1
Channel bandwidth	1.728 MHz
Simulation area	radius of 700 m, 1.5 km ²
Number of DECT-2020 NR RD	1200
RD deployment	Uniformly random
Building penetration loss	20 dB (ITU-R)
Indoor/outdoor fraction	80% / 20%
Channel model, Sink-Node	Urban macro
Channel model, Node-Node	NIST D2D model
Number of node/sink antenna elements	1
RD noise figure	7 dB
RD antenna element gain	0 dBi
Thermal noise level	-174 dBm/Hz
Link performance	Based on [5]
Traffic model	Poisson arrival
Application data message size	169 bytes including MAC, DLC and CVG headers
Channel access	RACH with LBT
LBT period	25 symbols
Contention window size	min – 15, max – 1024
LBT threshold	-97 dBm
Minimum link quality above Sensitivity level	15 dB
Error correction	HARQ with 7 attempts
Sensitivity	-99.7 dBm (TS 103.636-2)
Tx power	19 dBm
TX Format	MCS 1

and increase contention within the radio range, since each of those packets also needs to be served by the queue.

V. CONCLUSION

The performance evaluation of DECT-2020 NR downlink routing indicates that the routing strategy has a noticeable

TABLE II
ROUTING STRATEGY, CAPACITY, AND DELAY

Routing method	Capacity	Mean Delay
Selective Flooding, Full Flooding	≈ 6 pps	≈ 0.42 Sec
1st-hop only source routing, after 1st-hop selective flooding.	≈ 18 pps	≈ 0.53 Sec
Up to 2-hop source routing	≈ 40 pps	≈ 0.5 Sec
Up to 3-hop source routing	≈ 70 pps	≈ 0.31 Sec
Up to 5-hop source routing	≈ 135 pps	≈ 0.06 Sec
Up to 8-hop source routing	≈ 140 pps	≈ 0.043 Sec
End-to-End deterministic	≈ 140/145 pps	≈ 0.040/0.42 Sec

Note: In this stable operating region the packet outage is below 1% for each random individual realization.

impact on mesh-network downlink capacity in dense multi-hop deployments. Selective Flooding is robust to topology changes, but it creates substantial forwarding redundancy. This limits the downlink throughput in our setup to about 6 pps before delay and outages rise sharply. In contrast, Selective Source Routing uses deterministic forwarding: the capacity increases to 18 pps when source routing is introduced for the first hop. When the number of hops using source routing is increased to five, the capacity reaches about 135 pps, and adding additional source-routing hops provides only limited gains; end-to-end source routing achieves about 145 pps. In all presented results, the packet-outage percentage stays below 1%, meeting the reliability requirement defined for IMT-2020, while the DECT-2020 NR system uses only a single DECT-2020 NR channel.

In our simulation, each random deployment produces an independent network topology, while the topology remains fixed within a single realization. Furthermore, system operation was limited to a single channel with a fixed MCS (MCS 1), which limits single-link performance in terms of both experienced interference and achievable link data rate. Future work should address performance under dynamic route changes, multichannel operation, dynamic MCS selection, combined uplink–downlink traffic, and a fair cross-technology comparison.

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